CMPT 413/713: Natural Language Processing

Language Models

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Adapted from slides from Anoop Sarkar, Danqi Chen and Karthik Narasimhan

What will this course cover?

Representations

- Embeddings
- Compositional
- Structured

Methods

- Rule based
- Statistical Methods
- Neural Models
- Dynamic programming
- Language Models

First two months of the course

Applications and Tasks

- Machine Translation
- Dialogue Agents
- Information Extraction
- Question Answering
- Grounding

Last month of the course

Consider

Today, in Vancouver, it is 76 F and red

vs

Today, in Vancouver, it is 76 F and sunny

- Both are grammatical
- But which is more likely?

Language Modeling

- We want to be able to estimate the probability of a sequence of words
  - How likely is a given phrase / sentence / paragraph / document?

Why is this useful?

Applications

- Predicting words is important in many situations
  - Machine translation

$$P(\text{a } \mathbf{smooth} \text{ finish}) > P(\text{a } \mathbf{flat} \text{ finish})$$

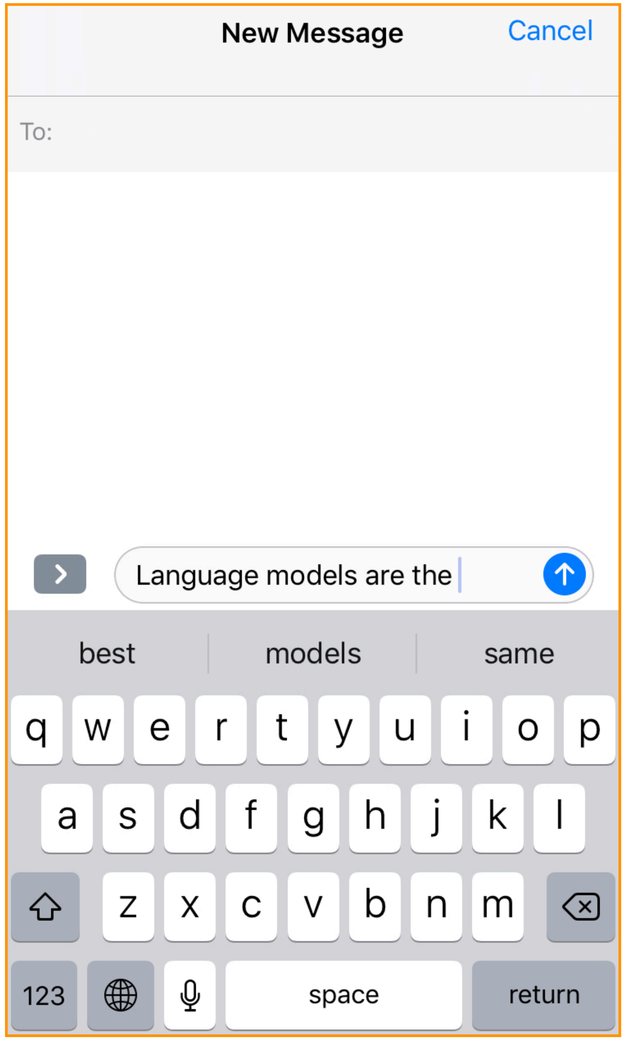
  - Speech recognition/Spell checking

$$P(\text{high school } \mathbf{principal}) > P(\text{high school } \mathbf{principle})$$

  - Information extraction, Question answering

Language models are everywhere

Autocomplete



Impact on downstream applications

Language Resources	Adaptation	Word	
		Cor.	Acc.
1. Doc-A		54.5%	45.1%
2. Trans-C(L)		63.3%	50.6%
3. Trans-B(L)		70.2%	60.3%
4. Trans-A(S)		70.4%	59.3%
5. Trans-B(L)+Trans-A(S)	CM	72.6%	63.9%
6. Trans-B(L)+Doc-A	KW	72.1%	64.2%
7. Trans-B(L)+Doc-A	KP	73.1%	65.6%
8. Trans-A(L)		75.2%	67.3%

PP
49972
1856.5
318.4
442.3
225.1
247.5
259.7
148.6

(Miki et al., 2006)

New Approach to Language Modeling Reduces Speech Recognition Errors by Up to 15%

December 13, 2018
Ankur Gandhe

Alexa | Alexa research | Alexa science

What we will cover today

- What is a language model?
  - Statistical language model using ngrams
- How to build a language model? Training the model from data
  Learning/estimating model parameters
  - MLE and smoothing
- How to use the language model? Generation
- How to tell if our language model is working well? Evaluation

What is a language model?

Probabilistic model of a sequence of words

Setup: Assume a finite vocabulary of words V

$$V = \{\text{cat, clown, crazy, killer, mat, on, sat, the}\}$$

V can be used to construct a infinite set of sentences (sequences of words)

$$V^+ = \{\text{clown, cat sat, killer clown, crazy clown, crazy cat,}$$
$$\text{crazy killer clown, killer crazy clown, } \ldots\}$$

where a sentence is defined as $s \in V^+$ where $s = \{w_1, \ldots, w_n\}$

What is a language model?

Probabilistic model of a sequence of words

Given a training data set of example sentences

$$S = \{s_1, s_2, \ldots, s_m\}, s_i \in V^+$$

Estimate a probability model

$$\sum_{s_j \in V^+} P(s_j) = \sum_j P(w_1, \ldots, w_{n_j}) = 1.0$$

Language Model

- p(clown) = 1e-5
- p(killer) = 1e-6
- p(killer clown) = 1e-12
- p(crazy killer clown) = 1e-21
- p(crazy killer clown killer) = 1e-110
- p(crazy slow killer killer) = 1e-127

Where do we get the vocabulary?

Common Setup: Assume a finite vocabulary of words V

- Get from a list of words (say a dictionary)
- Build from training data
  - Decide on vocabulary size (say $|V| = 50K$) and then pick most frequent words
  - Take words that occur more than T times.

Learning language models

How to estimate the probability of a sentence?

- We can directly count using a training data set of sentences
  - $P(w_1, \ldots, w_n) = \frac{C(w_1, \ldots, w_n)}{N}$
  - C is a function that counts how many times each sentence occurs
  - N is the sum over all possible $C(\cdot)$ values

Learning language models

How to estimate the probability of a sentence?

$$P(w_1, \ldots, w_n) = \frac{c(w_1, \ldots, w_n)}{N}$$

- **Problem**: does not generalize to new sentences unseen in the training data

  - What are the chances you will see a sentence?

    crazy killer clown crazy killer

  - In NLP applications, we often need to assign non-zero probability to previously unseen sentences

Estimating joint probabilities with the chain rule

$$P(w_1, w_2, \ldots, w_n) = P(w_1)P(w_2|w_1)P(w_3|w_1, w_2) \times \ldots \times P(w_n|w_1, w_2, \ldots, w_{n-1})$$
$$= \prod_{i=1}^{n} P(w_i|w_1, \ldots, w_{i-1})$$

Example Sentence: “the cat sat on the mat”

$$P(\text{the cat sat on the mat}) = P(\text{the}) \times P(\text{cat}|\text{the}) \times P(\text{sat}|\text{the cat})$$
$$\times \text{P}(\text{on}|\text{the cat sat}) \times P(\text{the}|\text{the cat sat on})$$
$$\times P(\text{mat}|\text{the cat sat on the})$$

Estimating probabilities

Let's count again!

$$P(\text{sat}|\text{the cat}) = \frac{\text{count}(\text{the cat sat})}{\text{count}(\text{the cat})}$$

$$P(\text{on}|\text{the cat sat}) = \frac{\text{count}(\text{the cat sat on})}{\text{count}(\text{the cat sat})}$$

$$\vdots$$

Maximum likelihood estimate (MLE)

- With a vocabulary of size $|V|$
  - # sequences of length n: $|V|^n$

Way too many parameters!

  - Typical vocabulary ~ 50k words
  - even sentences of length ≤ 11 results in $\approx 4.9 \times 10^{51}$ sequences!
    (# of atoms in the earth $\approx 10^{50}$)

Markov assumption

- Use only the recent past to predict the next word
- Reduces the number of estimated parameters in exchange for modeling capacity
- 0th order

$$P(\text{mat}|\text{the cat sat on the}) \approx P(\text{mat})$$

- 1st order

$$P(\text{mat}|\text{the cat sat on the}) \approx P(\text{mat}|\text{the})$$

- 2nd order

$$P(\text{mat}|\text{the cat sat on the}) \approx P(\text{mat}|\text{on the})$$

k^{th} order Markov

- Consider only the last *k* words for context

$$P(w_i \mid w_1 w_2 \ldots w_{i-1}) \approx P(w_i \mid w_{i-k} \ldots w_{i-1})$$

which implies the probability of a sequence is:

$$P(w_1 w_2 \ldots w_n) \approx \prod_i P(w_i \mid w_{i-k} \ldots w_{i-1})$$

(k+1) gram

n-gram models

Unigram $$P(w_1, w_2, \ldots, w_n) \approx \prod_{i=1}^{n} P(w_i)$$

Bigram $$P(w_1, w_2, \ldots, w_n) \approx \prod_{i=1}^{n} P(w_i|w_{i-1})$$

Trigram $$P(w_1, w_2, \ldots, w_n) \approx \prod_{i=1}^{n} P(w_i|w_{i-2}, w_{i-1})$$

and 4-gram, 5-gram, and so on.

Larger the n, more accurate and better the language model (but also higher costs)

Caveat: Assuming infinite data!

Estimating n-gram probabilities

- Maximum likelihood estimate (MLE): Use counts from text corpus

Unigram $$P(w_i) = \frac{C(w_i)}{\sum_{w_i \in V} C(w_i)} = \frac{C(w_i)}{N}$$

Bigram $$P(w_i|w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$$

Trigram $$P(w_i|w_{i-1}, w_{i-2}) = \frac{C(w_{i-2}, w_{i-1}, w_i)}{C(w_{i-2}, w_{i-1})}$$

Can reuse counts for multiple estimations

Maximum Likelihood Estimation

We want to find the set of parameters $\hat{\theta}$ that maximize the probability of the training data

$$\hat{\theta} = \arg\max_{\theta \in \Theta} \hat{L}(\theta; \mathcal{D})$$

Parameters

$$\theta : \{p(w_i | w_1, \ldots, w_{i-1})\}$$

Corpus of N sentences

Using our model, we can estimate the probabilities of these sentences

Likelihood

$$\hat{L}_m(\theta; \mathcal{D}) = \prod_{j=1}^{m} P(w_1^{(j)}, \ldots, w_{n_j}^{(j)}) = \prod_{j=1}^{m} \prod_{i=1}^{n_j} P(w_i^{(j)} | w_1^{(j)}, \ldots, w_{n_j}^{(j)})$$

Likelihood function

- How likely it is to see the examples in the training data
- Function of the parameters you are using to model the probability
- Probability density over data samples (sentences)

Log-Likelihood

$$\hat{\ell}_m(\theta; \mathcal{D}) = \sum_{j=1}^{m} \log P(w_1^{(j)}, \ldots, w_{n_j}^{(j)}) = \sum_{j=1}^{m} \sum_{i=1}^{n_j} \log P(w_i^{(j)} | w_1^{(j)}, \ldots, w_{n_j}^{(j)})$$

Easier to work with (products to sums)
Numeric underflow less of a issue

Typical assumptions

- Samples are iid (independently and identically distributed)

Maximum Likelihood Estimation (for categorical distributions)

- Unigram: $P(w)$
- Probability: $P(w) \geq 0, \sum_{w \in V} P(w) = 1$
- MLE is the sample mean
- Optimize $P_{\mathrm{ML}}(w) = \arg\max_{P(w)} \sum_{j=1}^{N} \sum_{i=1}^{n_j} \log P(w_i^{(j)}) = \arg\max_{P(w)} \sum_{w \in V} C(w) \log P(w)$
- Solve using Lagrange Multipliers $g(\lambda, P(\cdot)) = \sum_{w \in V} C(w) \log P(w) - \lambda(\sum_{w \in V} P(w) - 1)$

$$P(w) = \frac{C(w)}{\lambda} \longrightarrow P(w) = \frac{C(w)}{\sum_{w \in V} C(w)} = \frac{C(w)}{N}$$